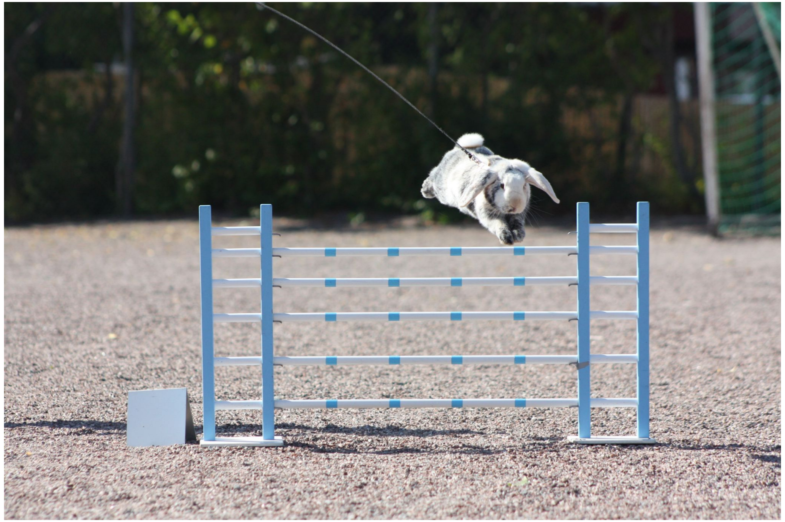
Multiple group item response theory in LEVANTE

LEVANTE Hackathon 2026

Mika Braginsky
Stanford University
mikabr@stanford.edu

Measurement in psychology is hard

How good is a rabbit at jumping?

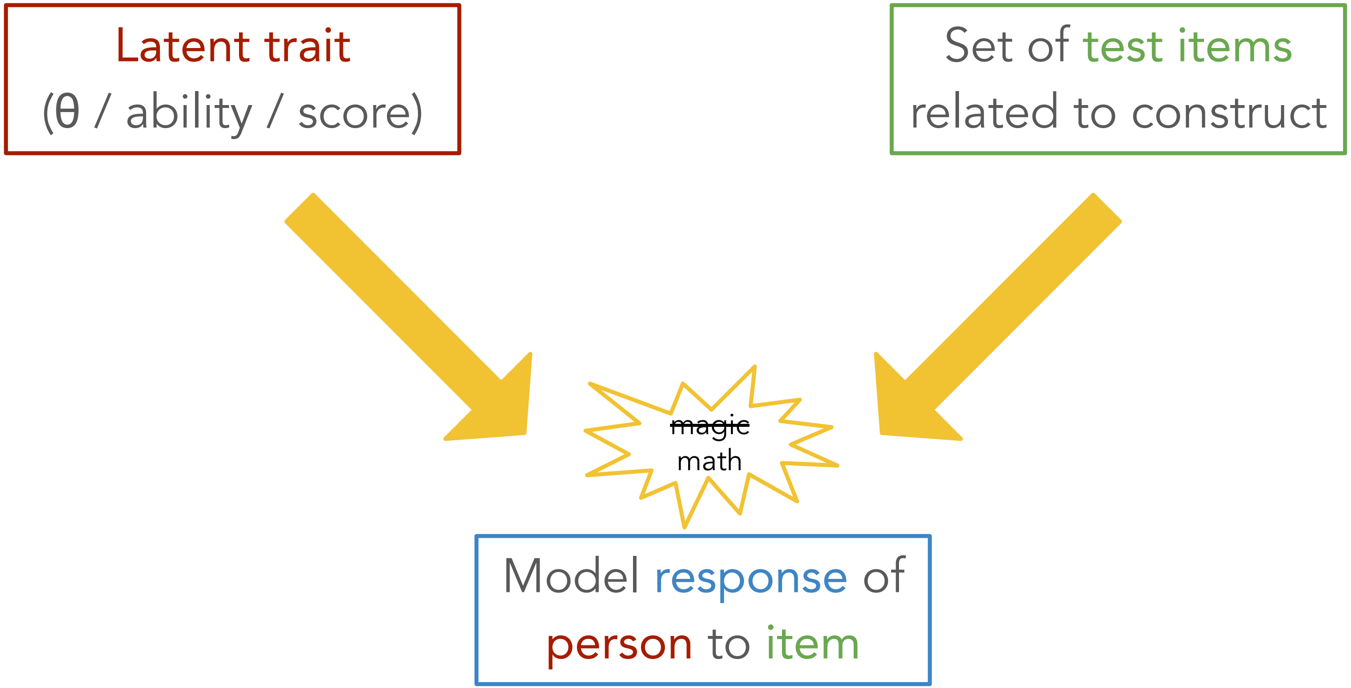


https://commons.wikimedia.org/wiki/File:Rabbit_jumping_over_a_fence_at_Rabbit_Show_Jumping.jpg

How good is a human at spatial reasoning?

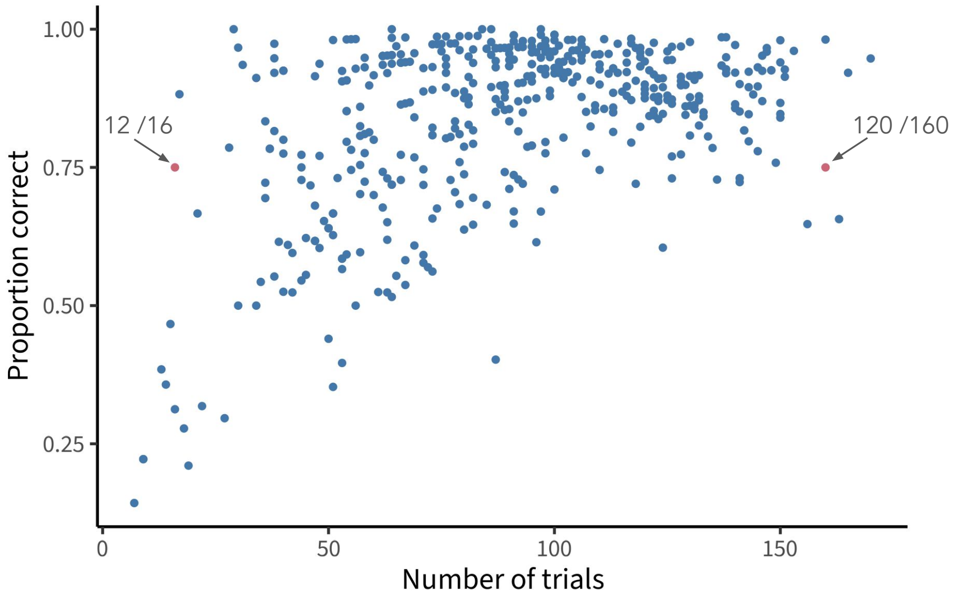
https://commons.wikimedia.org/wiki/File:A_young_child_engages_with_colorful_toys_on_a_table_in_a_bright_living_room.jpg

IRT building blocks

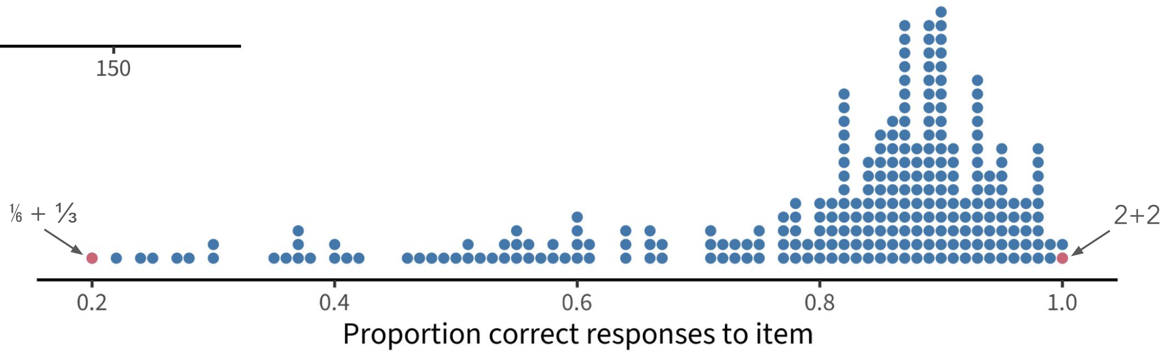


Accuracy is not enough

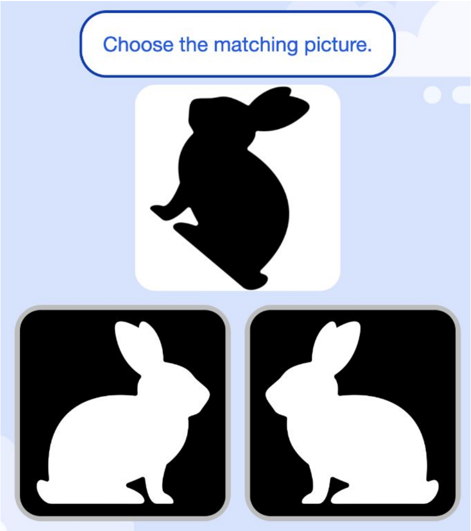
Proportions are not comparable

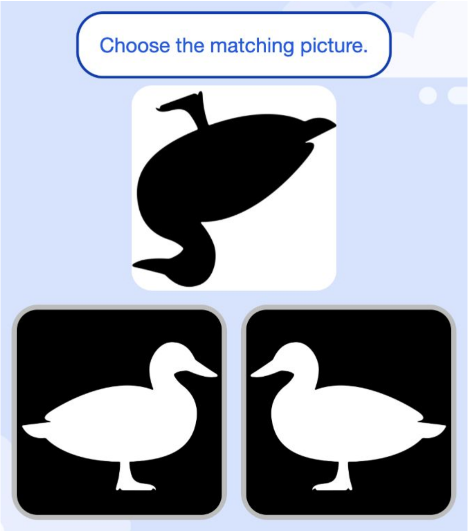


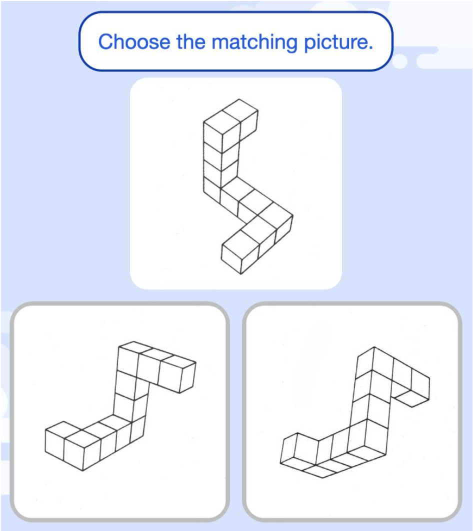
Items are not equivalent



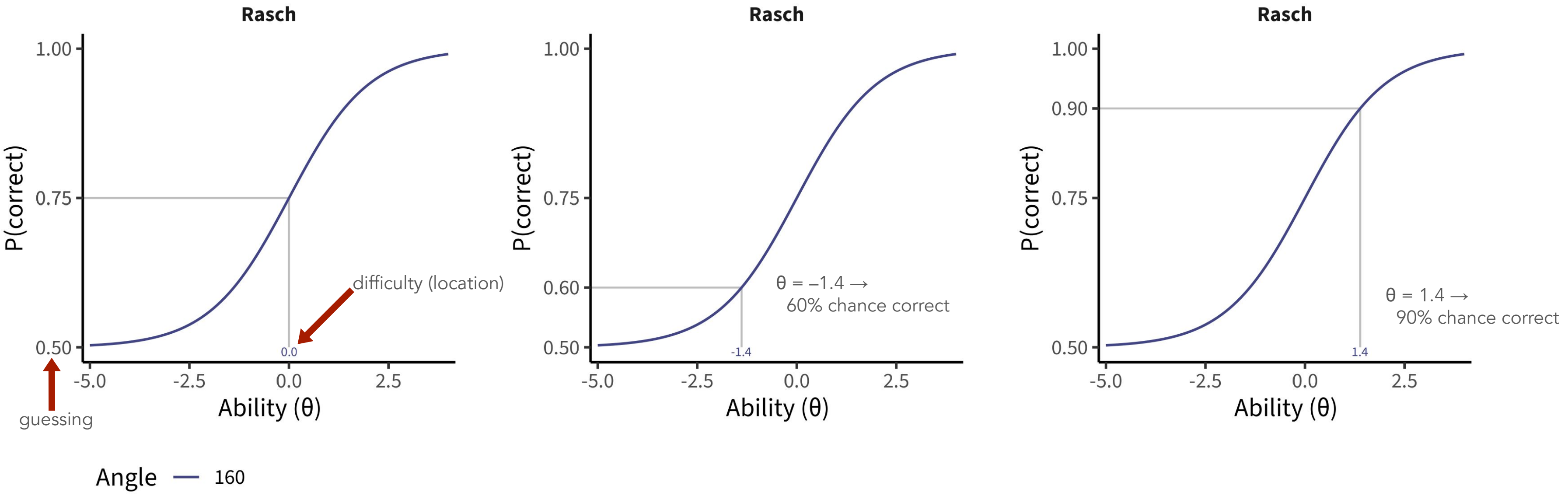
Shape rotation task

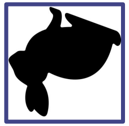




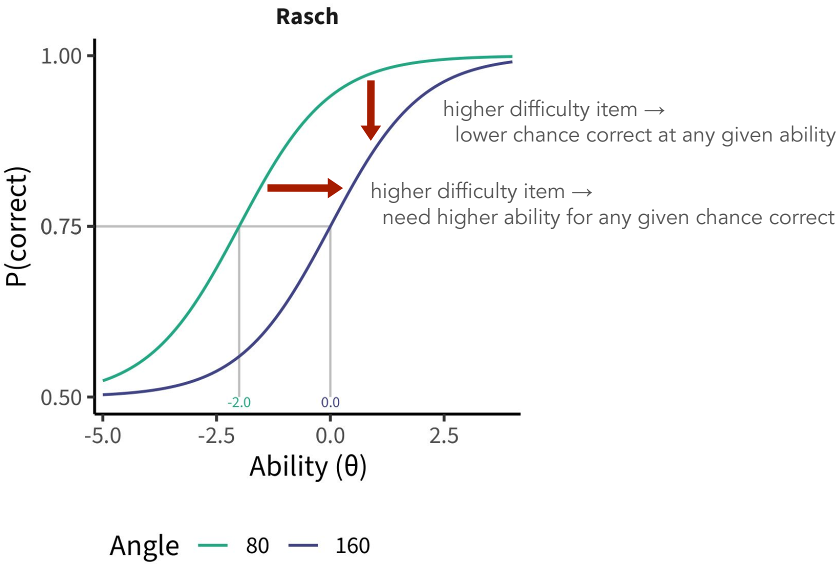


Rasch IRF

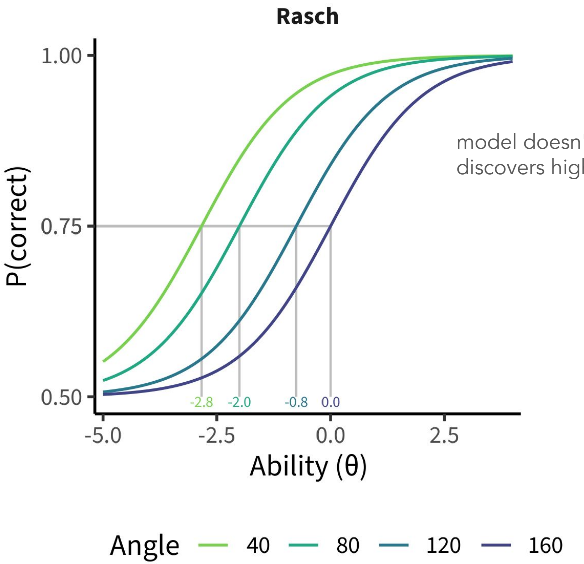

Rasch
1.00
0.75
0.50
P(correct)
-5.0
-2.5
0.0
2.5
Ability (θ)
0.0
difficulty (location)
guessing
Rasch
1.00
0.75
0.60
0.50
P(correct)
-1.4
θ = –1.4 →
60% chance correct
-5.0
-2.5
0.0
2.5
Ability (θ)
Rasch
1.00
0.90
0.75
0.50
P(correct)
1.4
θ = 1.4 →
90% chance correct
-5.0
-2.5
0.0
2.5
Ability (θ)
Angle
160



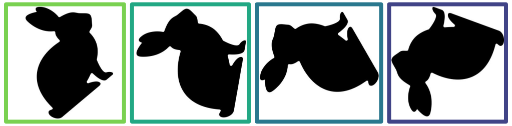
Rasch IRF


Rasch
1.00
0.75
0.50
P(correct)
higher difficulty item →
lower chance correct at any given ability
higher difficulty item →
need higher ability for any given chance correct
-2.0
0.0
-5.0
-2.5
0.0
2.5
Ability (θ)
Angle
80
160

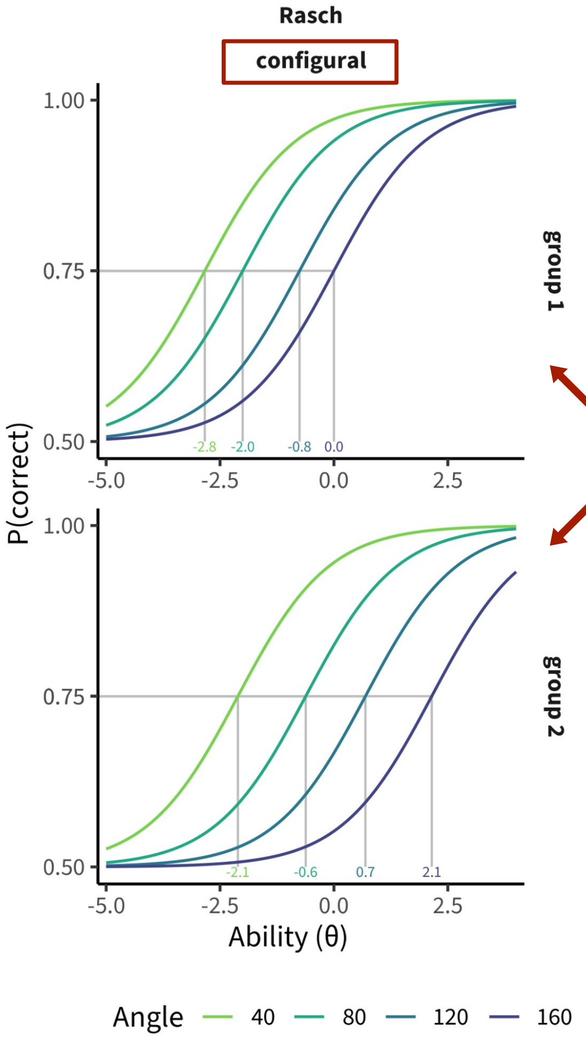
Rasch IRF



model doesn't know about angle but
discovers higher angle → more difficult

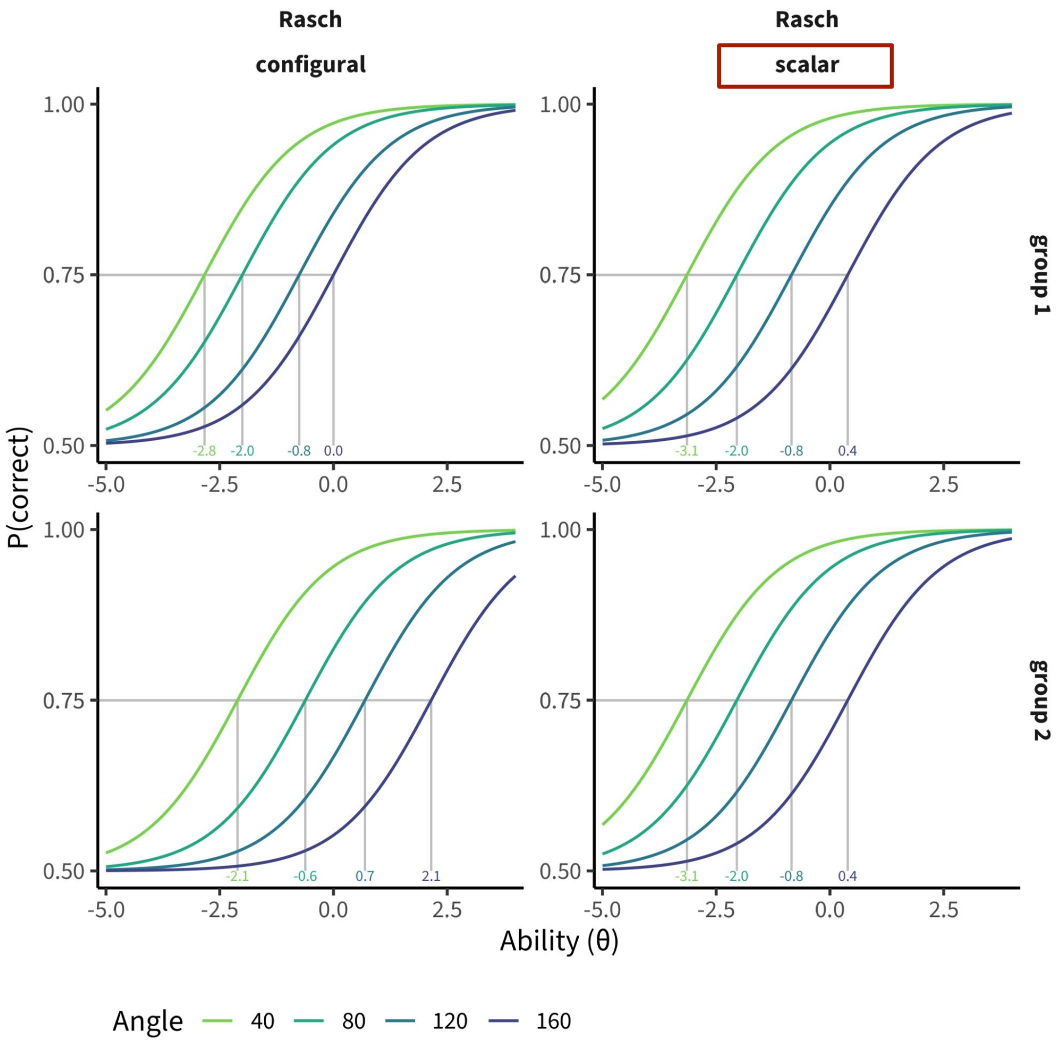


Rasch invariance



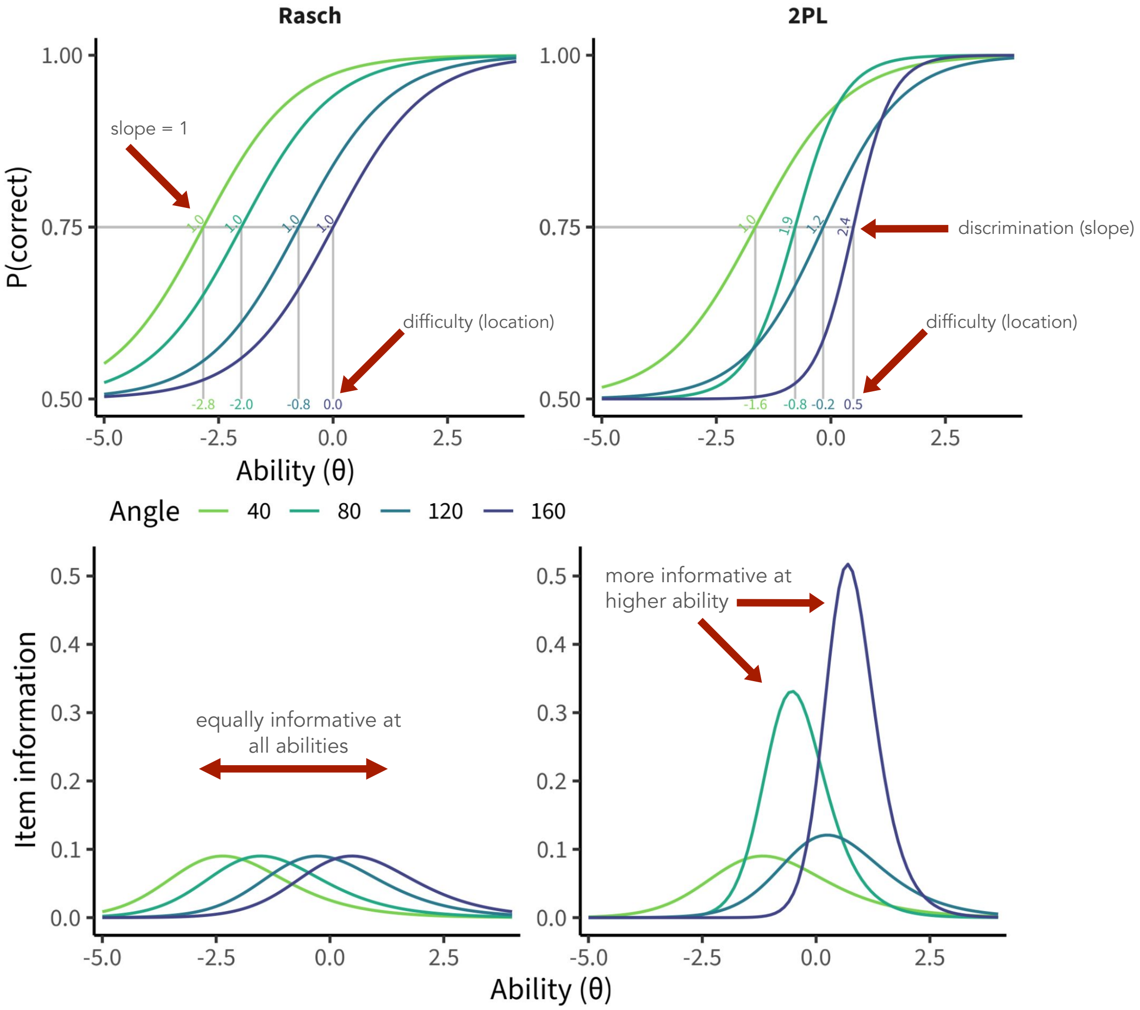
configural invariance →
groups have different difficulties

Rasch invariance



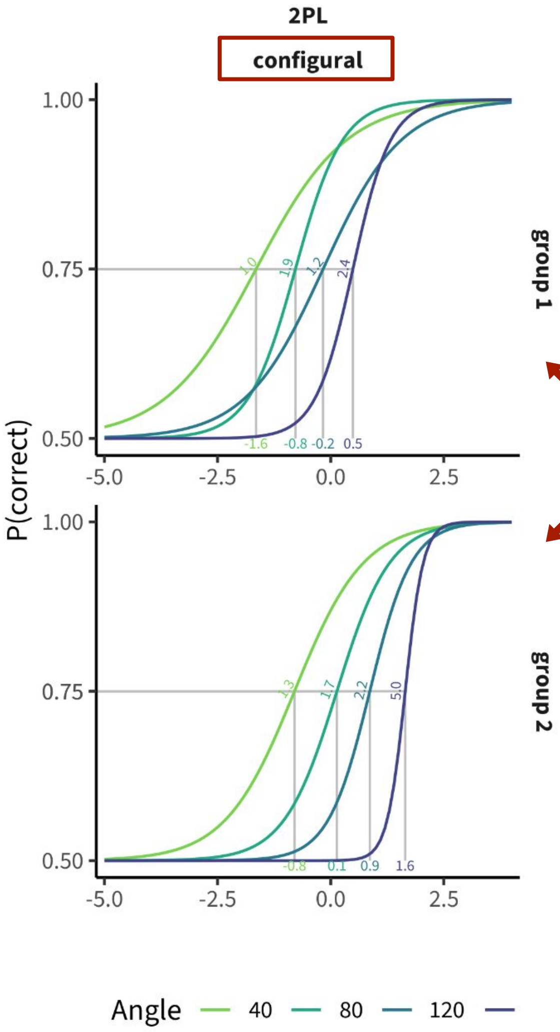
scalar invariance →
groups have same difficulties

Rasch vs 2PL



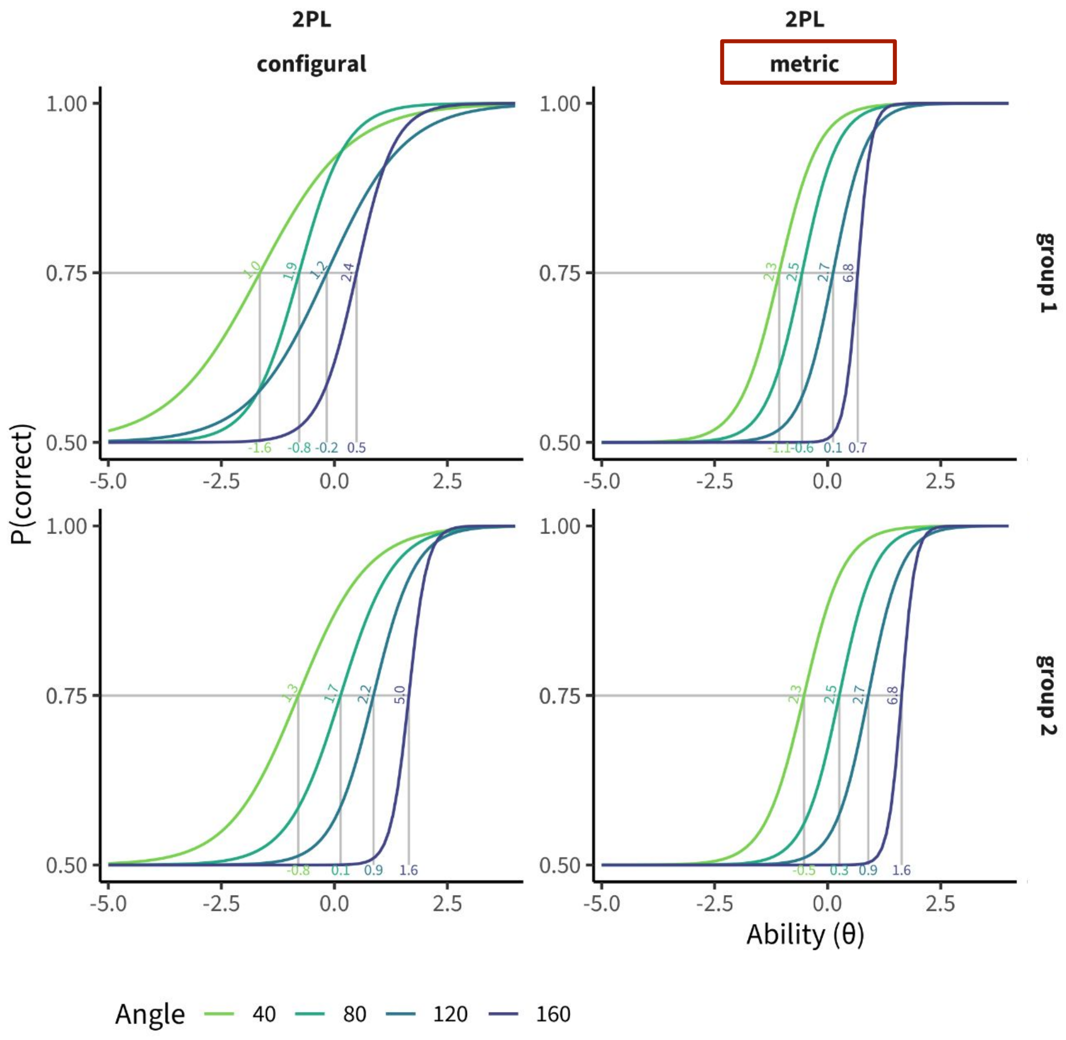
Rasch
2PL
P(correct)
slope = 1
difficulty (location)
discrimination (slope)
difficulty (location)
1.0
1.0
1.0
1.0
1.0
1.9
1.2
2.4
-2.8
-2.0
-0.8
0.0
-1.6
-0.8
-0.2
0.5
Ability (θ)
Angle 40 80 120 160
Item information
equally informative at all abilities
more informative at higher ability
Ability (θ)

2PL invariance



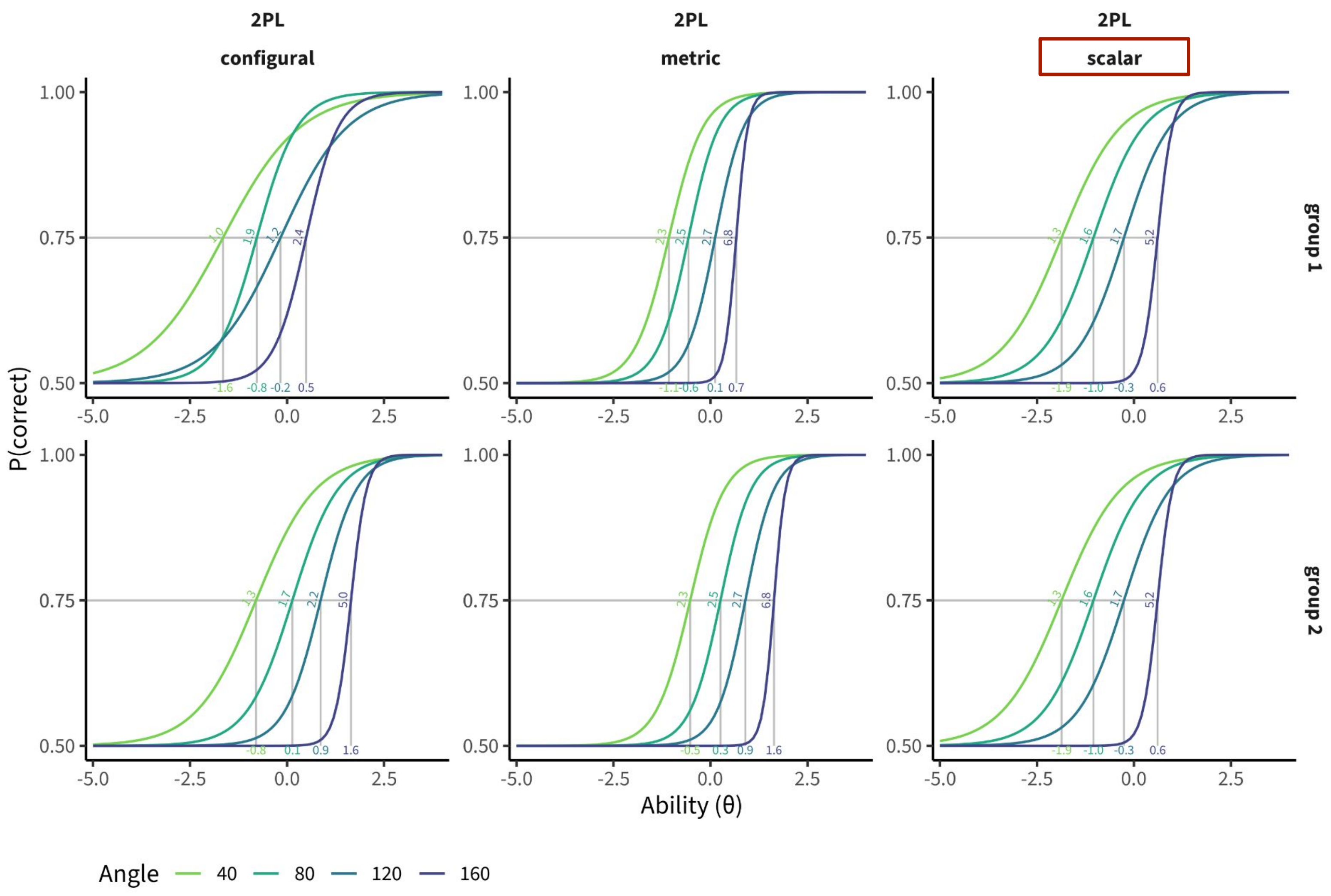
configural invariance →
groups have different difficulties and discriminations

2PL invariance



metric invariance →
groups have different difficulties, same discriminations

2PL invariance



scalar invariance → groups have same difficulties and discriminations

Measurement invariance

	Interpretation	Consequence
No invariance	Fractions are part of math in group 1, part of sentence understanding in group 2	Scores don't measure same construct across sites
Configural invariance	Fractions are easier in group 1 for a similar ability child than they are in group 2	Scores measure math ability within site but can't be compared across sites
Scalar invariance	Fractions are equally easy for a child of the same ability in group 1 and group 2	Scores are on the same scale and can be compared across sites

Model comparison

Bayesian Information Criterion (BIC)

how well does model fit data, without overfitting

Model	Invariance	BIC
Rasch	configural	53643
Rasch	scalar	53550
2PL	configural	53873
2PL	metric	53484
2PL	**scalar**	**53330**

Marginal reliability

how much variance in scores is captured by scale

Site	Rasch	2PL
mpieva-de	0.56	**0.90**
uniandes-co-bogota	0.53	**0.83**
uniandes-co-rural	0.50	**0.78**
western-ca	0.51	**0.81**

mirt: fitting models

IRT

```
mirt(
  data = df,
  itemtype = "Rasch",
  guess = 0.5,
  model = 1
)
```

columns = items
rows = observations
values = integers

"Rasch", "2PL", etc

single value or value per item

1 = unidimensional
can be string with constraints, priors, etc

Multigroup IRT

```
multipleGroup(
  data = df,
  itemtype = "Rasch",
  guess = 0.5,
  model = 1,
  group = df_groups,
  invariance = ""
)
```

string for each row

```
configural → ""
    metric → c("slopes")
    scalar → c("free_means", "free_var", "intercepts", "slopes")
```

mirt: extracting from models

parameters

```
coef(mod, simplify = TRUE)
```

easiness

```
coef(mod, simplify = TRUE,
     IRTpars = TRUE)
```

difficulty

scores

```
fscores(mod, full.scores.SE = TRUE)
```

various elements – see ?extract.mirt

```
extract.mirt(mod, "BIC")
```

marginal reliability (mirt)

```
marginal_rxx(mod)
```

marginal reliability (multipleGroup)

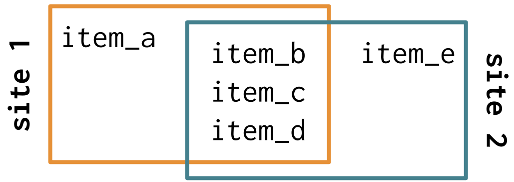
```
mod_g1 <- extract.group(mod, "group1")
marginal_rxx(mod_g1)
```

Complications

Problem: participants can see given item multiple times
Solution: de-duplicate items into instances and constrain instances to have equal parameters

id	item_uid	item_inst
1	item_a	item_a-1
1	item_a	item_a-2
1	item_b	item_b-1
1	item_b	item_b-2

Problem: items differ across sites
Solution: first fit model to overlapping items, then fit model to all items with parameters for overlapping items fixed to their value from first model



Thank you!